TRADEPULSE AI · INDEX OPTIONS · STRATEGY GUIDE

# VWAP-EMA Scalper

A 1-minute scalping strategy for NIFTY, BANKNIFTY and SENSEX options. It combines VWAP, the 5 and 9 EMAs, option delta, open interest, PCR, the premium's volume profile and India VIX. The premium is picked from the OI data. Risk-reward is set from delta and gamma, and trades are managed with a staged trailing stop.

| Timeframe | Criteria | Risk : Reward | Execution |
|---|---|---|---|
| **1-minute** | **7 (10 pts, all needed)** | **1:1 or 1:2** | **Paper (Scalper)** |

**Status and scope**

This document describes the strategy as implemented in the TradePlusAI repository (branch `claude/busy-planck-kqqyyn`). It has been tested on synthetic 1-minute data and has not yet run against live Groww market data. Automatic placement goes to the paper Smart Scalper only; live orders still need TradePulse's preview-and-PIN flow. This is software documentation, not investment advice.

Version 1.0 · 1 November 2026 · Source: `quant/src/tradepulse_quant/algo_engine/scalp_strategy.py`, `services/paper-trading/src/smart-scalper.ts`, `apps/web/components/scalper/`

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# 1. The strategy at a glance

The VWAP-EMA Scalper buys an index option (a **CE** for a long, a **PE** for a short) when the 1-minute index tape and the option chain agree on a short, clean move. It enters only when **all seven criteria** pass, worth 10 points in total, and every V5 rule gate is green.

1. **Trend and location.** Price is on the trade side of a sloping session VWAP and is not stretched away from it.
2. **Momentum trigger.** The 5 EMA leads the 9 EMA, price pulls back to the 9 EMA, and a candle resumes the move on above-average futures volume.
3. **Option confirmation.** The selected strike has a delta of 0.40-0.60. Option writers (OI) and the PCR support the side. The premium itself is accepted above its point of control. India VIX is in a tradeable range.
4. **Plan.** The stop is placed on the index chart (structure), then translated to the premium with **delta and gamma**. The target is the chosen 1:1 or 1:2 multiple. A trade is refused if charges eat more than a quarter of the reward.
5. **Execution and management.** The trade is placed on the paper Smart Scalper, automatically or with one click. The stop then moves to a cost-covered breakeven, locks profit, and trails.

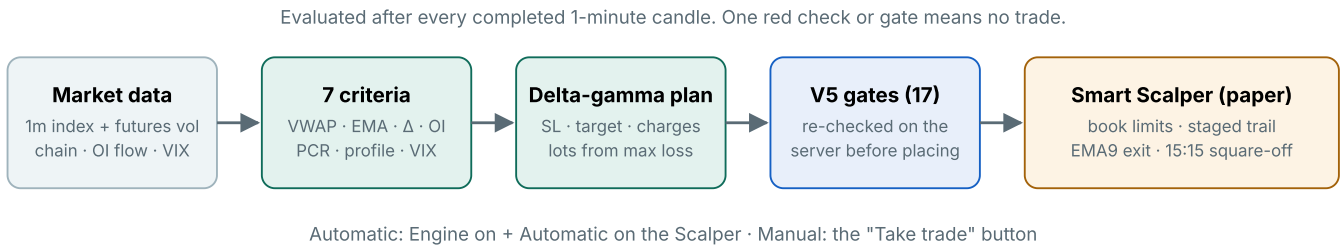



Figure 1. Decision pipeline. Each stage must pass before the next runs.

# 2. Data the strategy reads

All inputs are fetched server-side by `/api/algo-trading` and passed to the Python engine. If an input is missing, the criteria that need it **fail closed**: the strategy waits instead of guessing.

| Input | Source in TradePulse | Used for |
|---|---|---|
| Index 1-minute candles, today | Groww history, `timeframe=1m` | VWAP, 5/9 EMA, ATR(14), trigger, structural stop |
| Near-month futures volume | Attached to the index candles (`volume=futures`) | VWAP weighting and the trigger's volume check. Index candles carry no volume of their own. |
| Option chain (full, ±3%) | `/api/option-chain?full=1` | Premium, delta, IV, theta, OI, volume, lot and tick size for every strike |
| Options flow | Market-intel engine (`options_flow`) | Per-strike 5-minute OI change and build-up type, OI direction score, PCR and its 5-minute change, OI walls |
| India VIX | Market-intel volatility (value and regime) | Volatility criterion |
| Selected premium, 1-minute candles | `/api/market-data/option-history` (second pass) | The premium's volume profile (POC, VAH, VAL) |
| Operational state | Broker health, daily risk, contract master, SAFE_MODE, kill switch | V5 operational gates |

**Completed candles only**

The engine drops any candle that is still forming before it evaluates. This applies to the index (1-minute buckets) and to the premium. A signal can therefore never repaint, and it never confirms on a half-built candle. The page re-runs the engine 4 seconds after every minute closes.

**Two-pass evaluation**

Pass 1 reads the tape and picks the strike from the OI data. If that strike's premium candles are not loaded yet, the engine returns `needs_premium_candles`. Pass 2 adds those candles so the volume-profile criterion can be scored.

# 3. Choosing the side

The side comes from where price sits relative to VWAP and how the two EMAs are stacked, on the latest completed candle:

| Condition | Side | Buys |
|---|---|---|
| Close > VWAP **and** 5 EMA > 9 EMA | LONG | CE (call) |
| Close < VWAP **and** 5 EMA < 9 EMA | SHORT | PE (put) |
| VWAP and EMAs disagree | Lean only (the side of VWAP) | Nothing until both agree |
| Close exactly on VWAP | None | Nothing |

Only option buying is used. The PE side is the exact mirror of the CE side throughout this document: above becomes below, rising becomes falling, highs become lows.

# 4. The seven criteria

Each criterion scores points. The strategy trades only at **10 / 10**, so every criterion is effectively mandatory. The score is shown so a trader can see how close a setup is. The rules below are for a long (CE); the short (PE) side mirrors them.

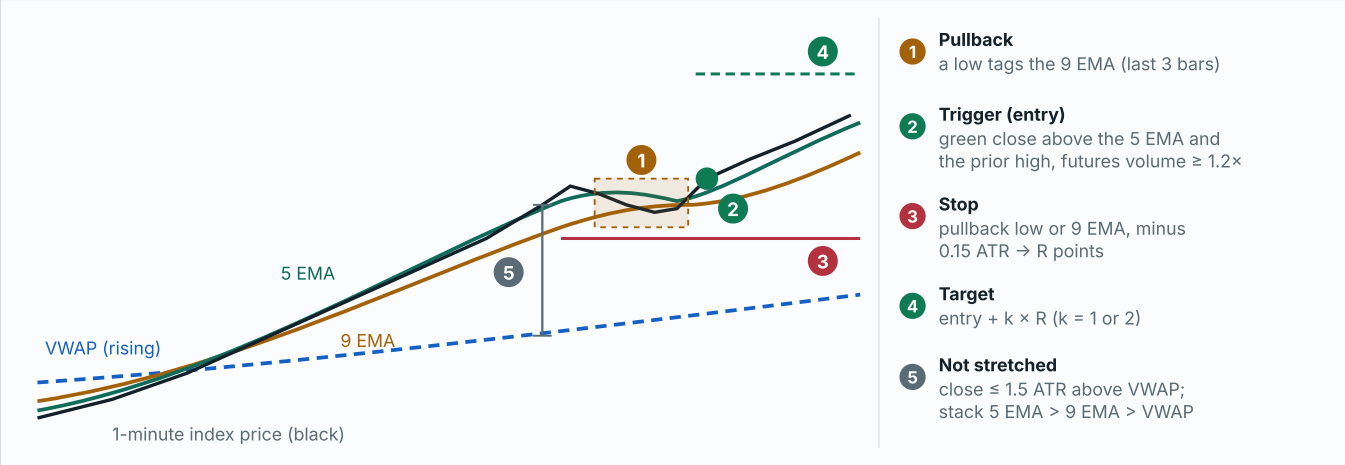



Figure 2. The long setup on the 1-minute index chart (schematic): trend above a rising VWAP, the EMA stack, the pullback to the 9 EMA, the resumption trigger, and the structural stop and target.

## ① VWAP

2 points

- **1 point:** the close is above the session VWAP **and** not more than **1.5 × ATR(1m)** away from it.
- **1 point:** VWAP has risen over the last **5 bars**.
- VWAP is weighted by near-month **futures volume**. Without volume it can't be trusted, so the criterion fails ("VWAP unobservable").

Why: VWAP is the session's fair price, where institutions benchmark fills. A long above a rising VWAP is with the session's flow. Stretching far above it is chasing, and those moves often snap back to VWAP inside a scalp's time frame.

## ② 5 / 9 EMA

2 points

- **1 point, the stack:** 5 EMA > 9 EMA > VWAP.
- **1 point, the trigger:** within the last 3 bars, a candle's low tagged the 9 EMA (tolerance 0.05 ATR). Then a candle that is **green**, closes **above the 5 EMA** and **above the previous candle's high**, on futures volume of at least **1.2×** the prior 20-bar average.
- **Freshness:** the trigger must be the latest completed candle, or the one before it if price hasn't run more than 0.5 ATR past the trigger close and is still above the 5 EMA.

Why: the 5/9 EMA pair is the classic fast momentum read on 1-minute charts. Buying the first resumption after a shallow pullback to the 9 EMA gives a defined stop just under the pullback. The volume check filters out drifts that nobody is trading.

## ③ Delta

1 point

- The auto-selected strike's **|delta| is between 0.40 and 0.60**, near the money.

Why: a scalp needs a premium that moves with the index point for point. Below 0.40 (OTM) the premium barely reacts and theta decay dominates. Above 0.60 (deep ITM) the option is a costly proxy for the future, with less leverage per rupee.

### ④ Open interest — 2 points

- **1 point:** the 5-minute OI writer-flow score is **+1 or more**. Put writing or call short covering is outweighing call writing.
- **1 point:** at the ATM strike there is **supportive build-up**: put short build-up (put writing), call short covering, or call long build-up. There must also be **no fresh call writing** at the ATM, and the **largest call-OI wall** above spot must be farther away than the target (in index points).

Why: option writers are the best-capitalised participants. Put writers defend support; call writers cap rallies. A long needs writers leaning its way and no big call wall sitting between the entry and the target.

### ⑤ PCR (put-call ratio, OI) — 1 point

- Long: PCR **0.80 to 1.60** and not falling over 5 minutes. Short: PCR **0.50 to 1.20** and not rising.
- Outside **0.50-1.60** the PCR counts as **stretched**, which signals reversal risk, and the criterion fails on either side.

Why: a healthy, rising PCR means puts are being written under the market (support). Extreme readings mean positioning is crowded and tends to unwind.

### ⑥ Volume profile of the selected premium — 1 point

- Built from the **selected premium's own** 1-minute candles today: the point of control (POC) and the 70% value area (VAL to VAH).
- Passes when the premium closes **above its POC and** either at or above the **VAH**, or on a **volume burst of at least 1.5×** the prior 20 bars.

Why: the option must confirm the index. A premium accepted above the price where most of today's contracts traded shows buyers in control of that contract, not just the underlying.

### ⑦ India VIX — 1 point

- VIX regime LOW, NORMAL or HIGH (not EXTREME) **and** a VIX value between **10 and 24**.

Why: below about 10, premiums hardly move and charges dominate. In extreme volatility, 1-minute structure breaks down, spreads widen and stops get run.

# 5. Auto-selecting the premium from OI

The strategy picks the contract itself; the trader doesn't choose a strike. Candidates are the trade side's options (calls for a long) with:

- |delta| between **0.35 and 0.65**. This is a wider net than the 0.40-0.60 the delta criterion requires, so near-misses can be shown.
- A premium of at least **₹20**. Where a bid/ask is available, a spread of at most **3%**; wider spreads are penalised.

Each candidate is ranked by a selection score:

```
score = 4 × (OI rank + volume rank) / (2 × (n - 1))   ← liquidity, 0-4
      + 2   if the option's 5-minute activity is LONG_BUILDUP or SHORT_COVERING
      - 2   if it is SHORT_BUILDUP (fresh writing of that option)
      + 2 × (1 - min(|delta - 0.5| / 0.15, 1))   ← closeness to delta 0.50, 0-2
      - 3   if the spread is wider than 3%
```

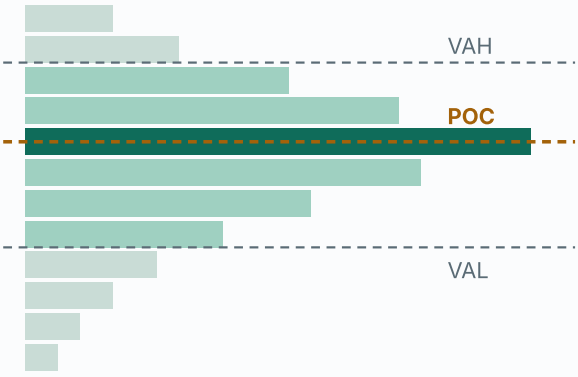
The top-ranked strike becomes the **selected premium**. On the Scalper desk the trade side's card and chart switch to it, marked with an OI pick badge. The ▲▼ buttons still override it until the next new pick. The next three candidates are listed with their scores.

# 6. The premium's volume profile

Each 1-minute premium candle's volume is spread evenly across the price bins its high-low range covers, using 24 bins over the day's range.

- **POC**: the bin with the most volume, where most contracts changed hands.
- **Value area**: grown outward from the POC, always adding the busier neighbouring bin, until it holds **70%** of the volume. Its top is the **VAH** and its bottom the **VAL**.

The criterion asks whether the premium is **accepted above value**: a close above the POC, plus either a break of the VAH or a 1.5× volume burst. The chart draws the POC, VAH and VAL as dashed lines.



volume at price → (70% value area shaded)

# 7. Stop, target and delta-gamma risk-reward

## 7.1 Structure first: the stop on the index

```
stop level   = min(pullback low, 9 EMA) - 0.15 × ATR(1m)        (long)
R (points)   = entry close - stop level
if R < 0.30 × ATR  → widen R to 0.30 × ATR     (tighter is inside 1-minute noise)
if R > 1.50 × ATR  → NO TRADE                   (too wide to be a scalp)
target level = entry + k × R,   k = 1 (1:1) or k = 2 (1:2)
```

The stop is where the scalp is proven wrong: back through the pullback and the 9 EMA. The target is a multiple of that risk, so reward-to-risk is fixed by design, not hoped for.

## 7.2 Mapping index points to the premium with delta and gamma

An option premium does not move one-for-one with the index. For a small index move *x*, its change is approximately:

$$\Delta\text{premium}(x) \approx |\text{delta}| \cdot x + \tfrac{1}{2} \cdot \text{gamma} \cdot x^2$$

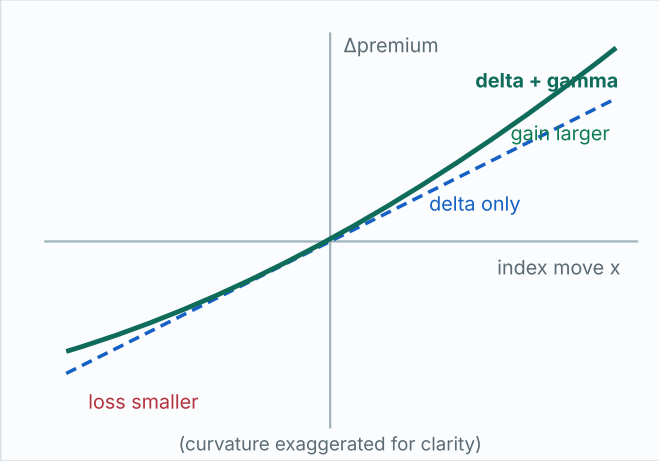
Applied to the stop and the target, with theta charged for the expected 10-minute hold:

```
theta drag    = |theta per day| × 10 / 375            (375 trading minutes a day)
premium SL    = P - ( |delta| · R  - ½ · gamma · R²  + theta drag )
premium target = P + ( |delta| · kR + ½ · gamma · (kR)² - theta drag )
P = entry premium (the ask);  delta from the chain;  gamma from Black-Scholes using the
chain's IV and expiry
```

**Why gamma matters.** Gamma makes the premium's response convex. On an adverse move the long option loses a little *less* than delta alone predicts ($-\frac{1}{2}\gamma R^2$). On a favourable move it gains a little *more* ($+\frac{1}{2}\gamma(kR)^2$). The premium reward-to-risk is therefore usually slightly better than the index ratio. Theta and tick rounding take some of that back.

**Effect by expiry.** With a week or more to expiry, gamma is small and the correction is a few paise. On expiry day gamma is large, so premium stops and targets move fast. The panel flags expiry days.

Both prices are rounded to the contract's tick (₹0.05).




## 7.3 The reward-to-risk gate

The V5 R:R gate compares the **premium** reward-to-risk with the ratio the trader chose, allowing 3% for tick rounding. Every other strategy in TradePulse keeps its 2.0 minimum.

| Chosen | Index target | Premium R:R needed |
|---|---|---|
| 1:2 (default) | 2 × R | ≥ 1.94 |
| 1:1 | 1 × R | ≥ 0.97 |

# 8. Charges and position size

Every plan is costed per lot, for both exits (target and stop), with a model tuned for liquid near-ATM index options:

| Component | Rate |
|---|---|
| Brokerage | ₹20 per order (₹40 round trip) |
| STT | 0.1% of the sell premium |
| Exchange transaction charge | 0.03503% of premium turnover |
| SEBI fee / stamp duty | ₹10 per crore / 0.003% of the buy premium |
| GST | 18% on brokerage + exchange + SEBI |
| Slippage | 0.15% per side, beyond the spread |
| Spread | The chain's bid/ask, capped at 0.5% of the premium per side (Groww's chain has no live bid/ask; the route estimates it from IV) |

> **The 25% rule**
> If round-trip charges at the target exceed **25% of the reward**, the trade is refused: the move is too small to scalp profitably. In practice a **1:1** NIFTY scalp with a 10-15 point stop is usually refused (about a ₹6 premium move, roughly a third of it lost to charges). **1:2** is the practical default; 1:1 passes when the structural move is larger, for example on BANKNIFTY.

**Lots.** lots = ⌊ max loss per trade ÷ (premium risk × lot size + charges at the stop) ⌋, capped by the Scalper's lots setting. The defaults are a ₹1,500 max loss and 1 lot, taken from the Scalper desk's own controls. If one lot already risks more than the max loss, there is no trade.

# 9. Worked example (real engine output)

Inputs: a synthetic NIFTY 1-minute tape (morning chop, a measured trend, a pullback to the 9 EMA, then a high-volume resumption candle at 09:40) and a synthetic chain and OI flow, run through the real engine path with every V5 gate. DEMO DATA, not live prices

| Criterion | Reading | Points |
|---|---|---|
| VWAP | Close 24,819.00 above VWAP 24,807.75, VWAP rising 2.4 pts in 5 min, 0.99 ATR from VWAP | 2 / 2 |
| 5/9 EMA | 5 EMA 24,816.23 > 9 EMA 24,815.17 > VWAP; pullback to the 9 EMA, resumed at 09:40 on 1.65× volume | 2 / 2 |
| Delta | NIFTY 24800 CE, delta 0.58 (picked from OI) | 1 / 1 |
| OI | Writer score +1; ATM 24800: call short covering, put short build-up; call wall 24950 is 131 pts away | 2 / 2 |
| PCR | 1.05, +0.02 in 5 min | 1 / 1 |
| Volume profile | Premium ₹125.40 above POC ₹125.01, on a 2.2× volume burst (VAH ₹126.70) | 1 / 1 |
| India VIX | 13.2, normal | 1 / 1 |
| **Total** | **All seven met; all 17 V5 gates green** | **10 / 10** |

| Step | Calculation | Result |
|---|---|---|
| Index stop / R | 24,819.00 − stop 24,807.29 | R = 11.71 pts |
| Index target (1:2) | 24,819.00 + 2 × 11.71 | 24,842.42 |
| Greeks | delta 0.576 · gamma 0.000791 · theta ₹9/day · IV 13.5% | theta drag ₹0.24 |
| Premium SL | ₹120.85 − (0.576 × 11.71 = ₹6.74 − gamma ₹0.05 + theta ₹0.24) | ₹113.90 |
| Premium target | ₹120.85 + (0.576 × 23.42 = ₹13.49 + gamma ₹0.22 − theta ₹0.24) | ₹134.30 |
| Premium R:R | (134.30 − 120.85) ÷ (120.85 − 113.90) | 1.94 (≥ 1.94 ✓) |
| Charges per lot | ₹139.92 at the target = 16% of the reward (≤ 25% ✓) | net R:R 1.25 |
| Size | ₹1,500 ÷ (6.95 × 65 + ₹136.05) → 2, capped at 1 lot | 1 lot = 65 qty |
| Max loss | 6.95 × 65 + charges at the stop | ₹587.80 |

**Same tape at 1:1:** target ₹127.40, charges 33% of the reward, net R:R 0.49. The trade is **refused** by the 25% rule.

# 10. The V5 rule gates

A confirmed strategy decision must also pass TradePulse's authoritative no-trade engine. One red gate blocks the trade. For this strategy the generic gates map as follows:

| Gate | For the VWAP-EMA scalper |
|---|---|
| Data quality | Valid, ordered candles; the latest under 15 minutes old during market hours |
| Session | Inside the entry window (09:35-14:45 IST; earlier cut-off on expiry day) |
| Option quote, daily risk, broker, contract metadata, reconciliation, execution | Operational state, supplied and verified on the server |
| Strategy | All seven criteria met and a valid plan |
| "Breakout" → **VWAP** | Criterion ① passed |
| "Retest" → **5/9 EMA trigger** | Criterion ② passed |
| VIX | Regime LOW / NORMAL / HIGH |
| Gap | Normal open, or a confirmed gap hold/fill |
| Score | 10 / 10 |
| Liquidity | The selected strike scores ≥ 2/3 (premium ≥ floor, OI > 0, traded today) |
| R:R | Premium R:R ≥ 0.97 (1:1) or ≥ 1.94 (1:2) |
| Charges vs reward | Charges ≤ 25% of the reward, and lots ≥ 1 |
| SAFE_MODE / kill switch | Either one blocks all new entries |

Not applied, by design: the 5-minute ADX regime gate (the 1-minute VWAP/EMA stack is the scalper's own trend filter) and the generic option-flow gate (OI and PCR are strategy criteria ④ and ⑤ instead).

# 11. Placing the trade on the Scalper

| Scalper setting | Behaviour |
|---|---|
| Strategy: **VWAP-EMA scalper** (default) | The Scalper trades this strategy. The level engine never enters, but it still manages exits. |
| Engine **on** + Trade **Automatic** | A confirmed setup is placed automatically, once per setup (keyed on the trigger candle and side). |
| Engine **on** + Trade **Manual** | "⚡ Take trade · BUY 24800 CE 1:2" lights up when confirmed; one click places it. |
| Strategy: Level scalper | The desk's original support/resistance engine. |

Every placement takes the same safe path:

1. **Server re-check.** `POST /api/algo-trading` with `action: PLACE_ON_SCALPER` re-runs the strategy and all V5 gates on the server. The browser can't confirm a trade.
2. **Book limits.** The Smart Scalper applies its own limits: kill switch, SAFE_MODE, daily loss limit, a 5-minute cool-down after a loss, a stop after 3 consecutive losses, the trade cap, one engine position per index, and 09:15-15:15 IST on trading days only.
3. **Fill.** The leg is filled at the live ask, plus the scalper's configured slippage. Slippage reduces the reward, never the stop distance.

> **Paper only**
> Automatic placement never sends a live order. Real-money orders remain on TradePulse's separate live path: `/api/live-orders`, with a preview, a PIN, the LIVE limits and the kill switch.

# 12. Trade management: SL, target and trailing SL

Once in, the position is managed by a **staged, cost-covered trail**. R is the initial premium risk (entry − initial SL). The stop only ever moves **up**.

| Stage | Triggered when the best premium reaches | SL moves to | Example (entry ₹120.85, R ₹6.95) |
|---|---|---|---|
| **Initial** | entry | the delta-gamma SL | ₹113.90 |
| **Breakeven** | +0.5R (₹124.33) | entry + round-trip charges per unit (cost-covered) | ₹123.00 |
| **Locked** | +1R (₹127.80) | entry + 0.5R | ≈ ₹124.32 |
| **Trailing** | beyond +1R | best premium − 0.6R | at ₹131 → ₹126.83 |

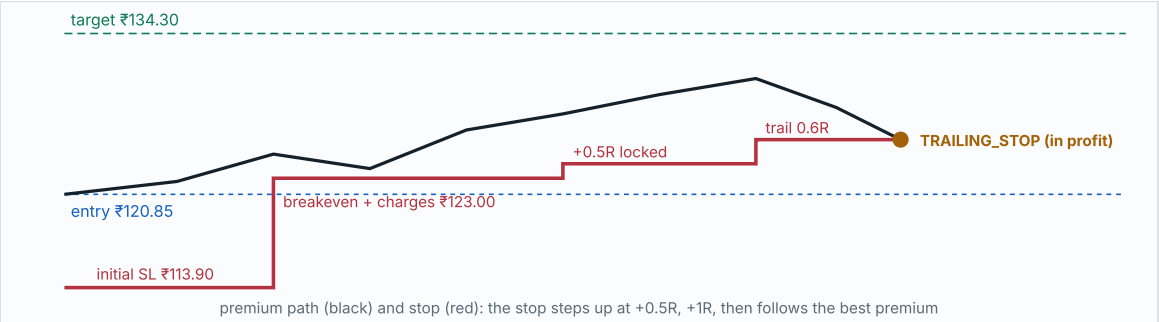



Figure 3. The staged trail on a winning scalp.

## Exits, in priority order

| Exit reason | Rule |
|---|---|
| `SQUARE_OFF_15_15` | Every open position is closed at 15:15 IST |
| `MAX_LOSS_PER_TRADE` | The open loss reaches the max loss per trade |
| `SPOT_LEVEL_BROKEN` / `SPOT_TARGET` | The index trades through the structural stop / reaches the index target |
| `TARGET` | The premium reaches the premium target |
| `STOP_LOSS`, `BREAKEVEN_STOP`, `LOCKED_PROFIT_STOP`, `TRAILING_STOP` | The premium reaches the stop; the reason names the trail stage |
| `EMA9_LOST` | After breakeven: a **completed** 1-minute index candle closes back below the 9 EMA (above it, for a PE). The scalp's reason is gone. |
| `SCALP_TIME_STOP` | The trade hasn't reached +0.5R within the Scalper's time stop (default 12 minutes) |

The trail stage, the initial SL and the charges per unit are saved with the order, so a server restart resumes the same trail. On the premium chart the risk-reward drawing follows the live SL. Once the SL is above the entry, it shows the profit locked, e.g. "SL · +0.5R locked ₹124.32 (+₹225.55)". **Keep the execution page open:** exits are evaluated by the Scalper desk's 10-second scan, which runs in the browser while the page is open, the same way the Smart Scalper has always worked.

# 13. Screens

All screens below show DEMO DATA: the built application rendering the real engine's output for the synthetic tape of section 9. They are not live market prices.

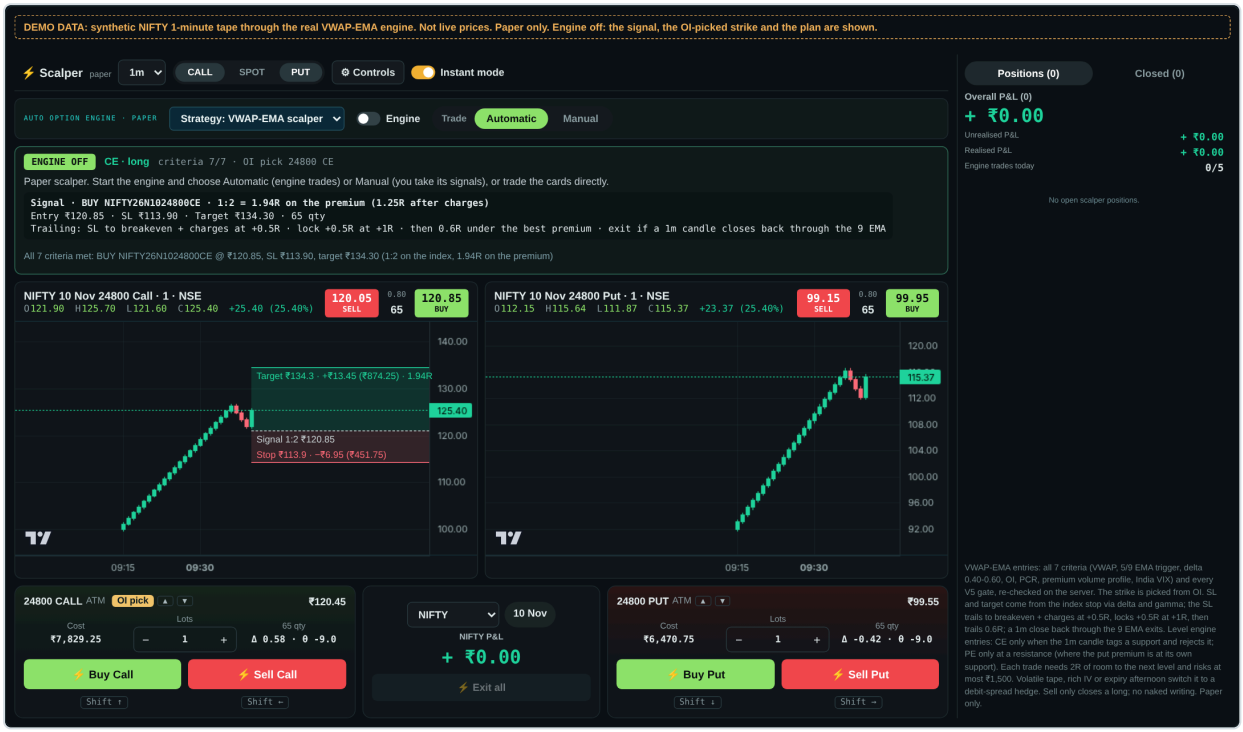



Figure 4. The Scalper desk with Strategy set to VWAP-EMA: criteria 7/7, the OI-picked 24800 CE (badge), the plan (entry ₹120.85, SL ₹113.90, target ₹134.30) and the risk-reward drawing on the call chart.

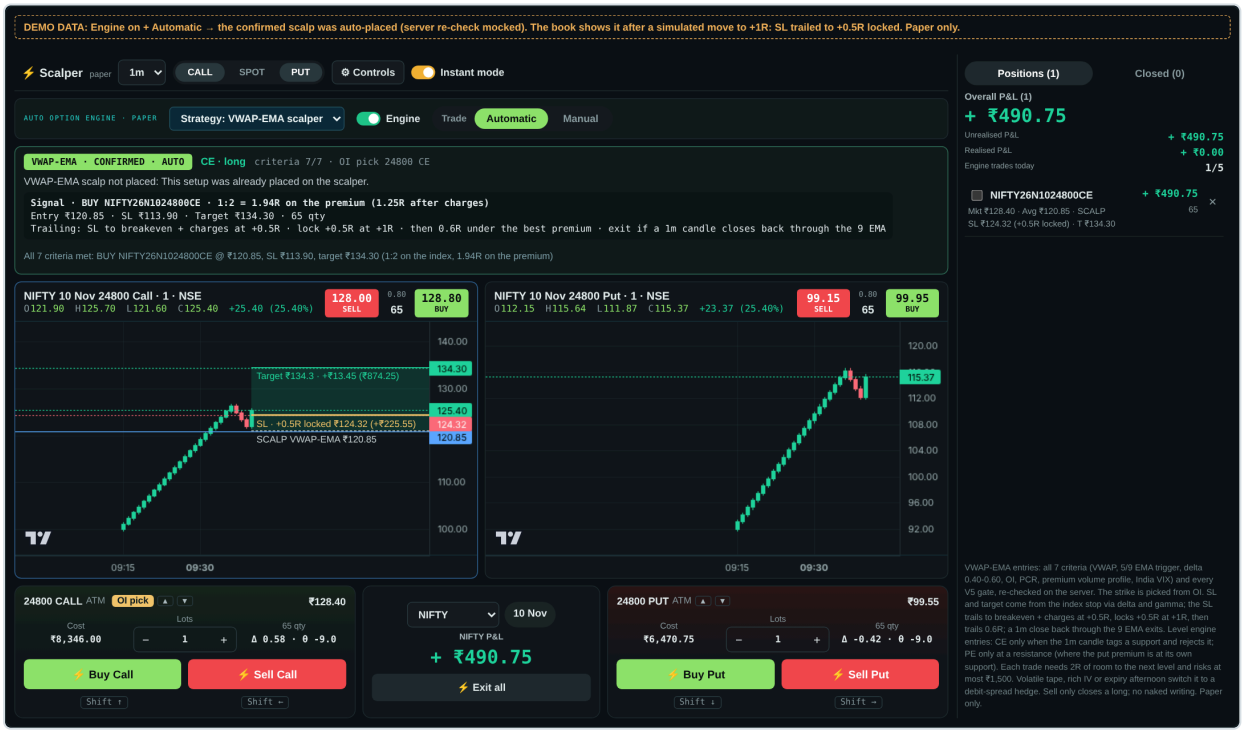



Figure 5. Engine on + Automatic: the scalp was placed (server re-check mocked for the demo). Shown after a simulated move to +1R: the SL has trailed to "+0.5R locked" and the position cannot lose. A second attempt on the same setup is refused as already placed.




Figure 6. The strategy panel on the execution page: the 1:1 / 1:2 selector, criteria tiles and reasons, the OI-selected premium with its Greeks, the delta-gamma working of the SL and target, the premium chart with POC / VAH / VAL and the risk-reward zones, and the 17 V5 gates.

# 14. Parameter reference

Defaults in `ScalpConfig` (Python) and `VWAP_EMA_TRAIL` (TypeScript). The max loss per trade and the lots cap come from the Scalper desk's Controls.

| Parameter | Value | Meaning |
|---|---|---|
| EMA fast / slow | 5 / 9 | Momentum pair on 1-minute closes |
| ATR period | 14 | 1-minute ATR used for every distance rule |
| Minimum session bars | 15 | No evaluation before 15 one-minute candles today |
| VWAP slope bars | 5 | VWAP must rise (fall) over this many bars |
| Max distance from VWAP | 1.5 ATR | Beyond this the entry is a chase |
| Pullback window / EMA tolerance | 3 bars / 0.05 ATR | Where the 9 EMA tag must occur |
| Trigger volume | ≥ 1.2× | Futures volume vs the prior 20 bars |
| Trigger age / max drift | ≤ 1 bar / 0.5 ATR | Freshness of the signal |
| Strike pool delta / criterion delta | 0.35-0.65 / 0.40-0.60 | Selection band / pass band |
| Min premium / max spread | ₹20 / 3% | Candidate filters |
| PCR long / short / stretched | ≥ 0.80 / ≤ 1.20 / <0.50 or >1.60 | PCR bands |
| India VIX | 10-24, not EXTREME | Tradeable volatility |
| Profile bins / value area / burst | 24 / 70% / 1.5× | Premium volume profile |
| Stop buffer / min R / max R | 0.15 / 0.30 / 1.50 ATR | Structural stop rules |
| Hold for theta | 10 min | Theta charged on the plan |
| Max charges share | 25% | Of the reward at the target |
| Slippage / spread cap | 0.15% / 0.5% per side | Scalper cost model |
| R:R tolerance | 3% | Tick-rounding allowance on the premium ratio |
| Trail: breakeven / lock / lock to / gap | +0.5R / +1R / +0.5R / 0.6R | Staged trailing stop |
| Max loss per trade / lots cap | ₹1,500 / 1 | Scalper Controls (defaults) |
| Scalp time stop | 12 min | Scalper Controls (default) |

# 15. Limitations and first-live checklist

- **Not yet run on live data.** The build environment could not reach market data. The strategy is covered by tests on synthetic tapes (21 Python and 14 TypeScript tests for the strategy, placement, trail and chart), and the full suites pass. Watch the first live sessions in paper mode.
- **Check on the first live sessions:**
  - near-month futures volume arrives on the 1-minute index candles, so VWAP is observable;
  - market-intel's per-strike 5-minute OI change and build-up are populated (they need a 5-minute baseline after the open);
  - the selected premium's 1-minute candles load for the volume profile;
  - the chain's delta and IV look sane on expiry day.
- **Spread estimate.** Groww's chain carries no live bid/ask, so the spread is estimated from IV and capped. Real fills may differ.

- **1:1 rarely qualifies on NIFTY** because of the 25% charges rule. That is by design, not a fault.
- **Browser-driven management.** Exits run while the execution page is open.
- **Paper first.** Run the strategy in paper mode for several weeks and review the closed-trade log (exit reasons, R multiples, charges) before considering live trading through TradePulse's PIN-confirmed flow.

> **Risk disclaimer**
> Options trading involves substantial risk, and most intraday traders lose money. This document describes software behaviour; it is not investment advice, and TradePulse is not a SEBI-registered adviser. Live API trading may require static-IP registration and algo approval with your broker under SEBI's retail algo framework.

# Appendix: code and tests

| File | Role |
|---|---|
| `quant/src/tradepulse_quant/algo_engine/scalp_strategy.py` | Criteria, strike selection, volume profile, delta-gamma plan, charges, sizing |
| `quant/src/tradepulse_quant/algo_engine/engine.py` | Wiring into the V5 engine (`_scalp_evidence`); completed-candle filtering |
| `quant/src/tradepulse_quant/algo_engine/no_trade_engine.py` | The V5 gates; strategy-specific R:R minimum |
| `apps/web/app/api/algo-trading/route.ts` | Inputs, two-pass evaluation, `PLACE_ON_SCALPER` with the server re-check |
| `services/paper-trading/src/smart-scalper.ts` | `place()`, `vwapEmaTrail()`, `emaMomentumLost()`, exits |
| `apps/web/components/scalper/scalper-desk.tsx` | Strategy selector, auto and manual placement, OI-pick strike, drawings |
| `apps/web/components/scalper/premium-chart.tsx` | Risk-reward drawing (reward zone, risk zone, locked-profit line) |
| `quant/tests/test_scalp_strategy.py` | Criteria, delta-gamma maths, 1:1 vs 1:2, PE mirror, V5 integration |
| `tests/unit/scalp-place.test.ts`, `tests/contract/algo-trading-route.test.ts` | Placement limits, staged trail, momentum exit, chart layout, route guards |
| `docs/operations/vwap-ema-scalper.md` | Operator notes (Markdown) |